

ICT461 — Unit 1 Lab Assignment Report

Course Registration Interface

Student Name: Dennis Sampa

Student ID: 202302571

Course: ICT461

GitHub Repository: <https://github.com/Ghost-0402/-ict461-unit1-lab>

1. Purpose

This report documents the design, development, and testing of a one-page course registration interface built for ICT461. The interface uses semantic HTML, responsive CSS, and a JavaScript interaction to provide a live registration summary. The report includes evidence of the working page, DevTools inspection, accessibility testing, and version control via Git and GitHub.

2. Working Page

The screenshot displays the 'ICT461 — Course Registration' interface. The header includes the title and subtitle 'Unit 1 Lab Assignment — Semester Registration Portal'. The main content is divided into two sections: 'Student Registration Form' and 'Registration Summary'.

Student Registration Form:

- Full Name:** Dennis Sampa
- Student Email:** sampadennis0402@gmail.com (with a note: 'Use your university email address.')
- Student ID:** 202302571 (with a note: '9-digit ID, e.g. 202345678.')
- Course:** ICT332 — Generative AI & LLMs (K2000)
- Study Mode:** Full-time (selected), Part-time
- ☒ I confirm the information above is correct and I agree to the registration terms.
- Register** button
- Success Message:** Registration successful! A confirmation has been sent to sampadennis0402@gmail.com.

Registration Summary:

- Name:** Dennis Sampa
- Course:** ICT332
- Study Mode:** Full-time
- Tuition Fee:** K2000
- This panel updates live as you fill in the form — no page reload required.

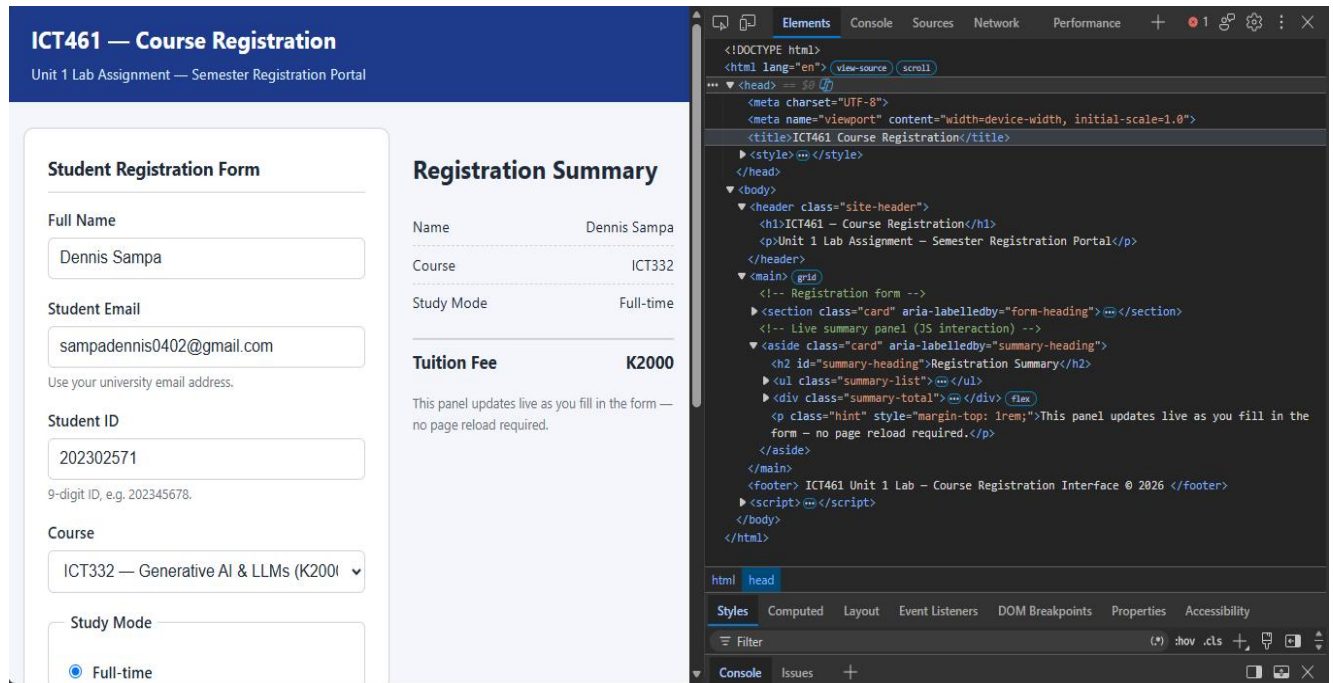
ICT461 Unit 1 Lab — Course Registration Interface © 2026

Purpose: This screenshot shows the finished registration form running in the browser, demonstrating that all fields, labels, and the live summary panel work as intended.

What I did: I opened the HTML file directly in Microsoft Edge and filled in each field (name, email, 9-digit student ID, course, and study mode) to confirm the form behaved correctly and that the Registration Summary panel updated in real time without reloading the page.

3. DevTools Evidence

3.1 Elements Panel



Purpose: This screenshot shows the semantic HTML structure of the page in the Elements panel, confirming the use of proper landmark elements and ARIA attributes.

What I did: I opened DevTools with F12, selected the Elements tab, and expanded the header, main, and footer elements to show the semantic structure, including the aria-labelledby attributes linking each section to its heading.

3.2 Network Panel

The screenshot displays a web browser window with the URL `C:/Users/HP/Downloads/ICT461_LAB1.html`. The page title is "ICT461 — Course Registration" and the subtitle is "Unit 1 Lab Assignment — Semester Registration Portal".

The page content is divided into two main sections:

- Student Registration Form:** Contains input fields for "Full Name", "Student Email" (with a note "Use your university email address."), "Student ID" (with a note "9-digit ID, e.g. 202345678."), "Course" (a dropdown menu showing "-- Select a course --"), and "Study Mode" (a radio button selected for "Full-time").
- Registration Summary:** Displays the following information:
 - Name: _____
 - Course: _____
 - Study Mode: Full-time
 - Tuition Fee: K0**
 - A note: "This panel updates live as you fill in the form — no page reload required."

The DevTools Network panel is open on the right side of the browser. It shows a single request for `ICT461_LAB1.html` with a status of 200, a type of document, and a size of 14.0 kB. The load time is 15 ms. The panel also shows a timeline of the request and a summary of the page load: 1 requests, 14.0 kB transferred, 14.0 kB resources, Finish: 15 ms, DOMContentLoaded: 99 ms, Load: 105 ms.

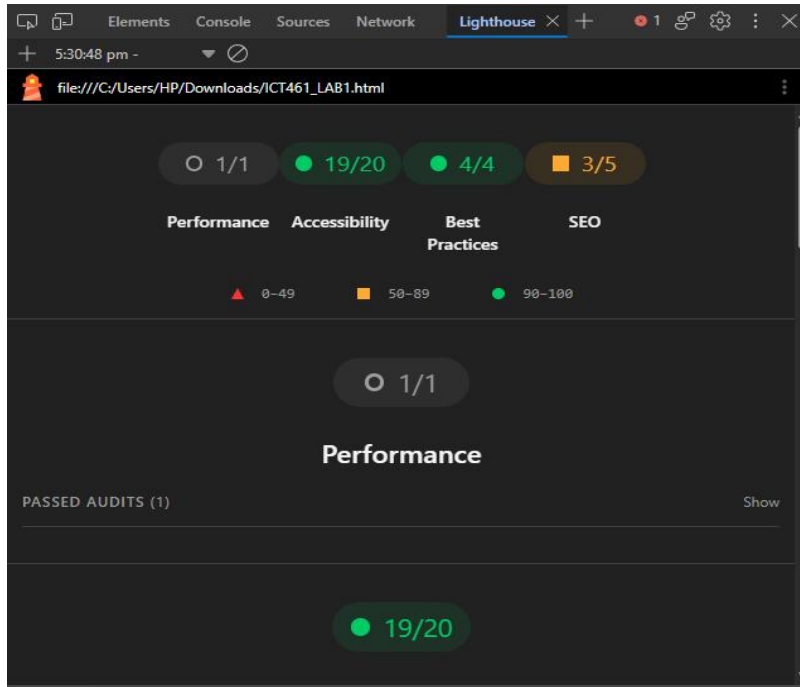
Name	Status	Type	Initiator	Size	Time	Fulfilled ...
ICT461_LAB1.html	200	document	Other	14.0 kB	15 ms	

Purpose: This screenshot shows the Network tab confirming the page loads successfully.

What I did: I opened the Network tab in DevTools and reloaded the page to capture the single HTML document request, which returned a 200 status and loaded in around 15ms.

4. Accessibility Checks

4.1 Initial Lighthouse Audit



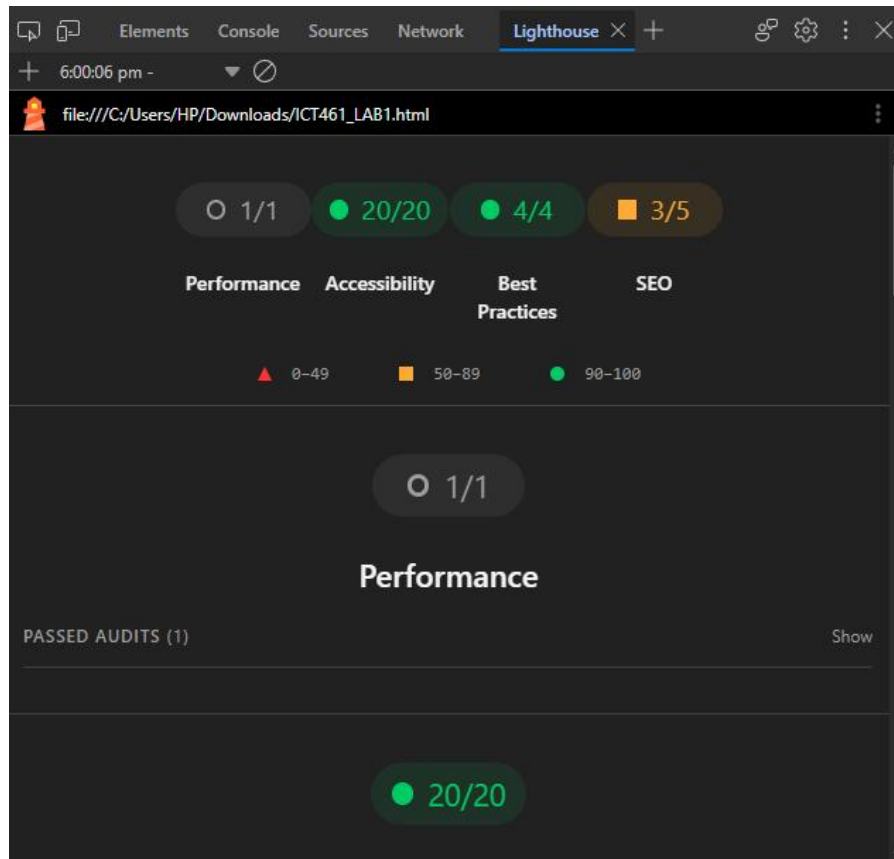
Purpose: I ran a Lighthouse accessibility audit to check the page against WCAG standards before making any further changes.

What I found: The audit returned a score of 19/20, with one failing check flagging insufficient colour contrast between the muted Grey text (used for the hints and footer) and its background.

4.2 Fix Applied

Changed the `--color-muted` CSS variable from `#6b7280` to `#4b5563` to increase the contrast ratio of hint text and footer text against the background.

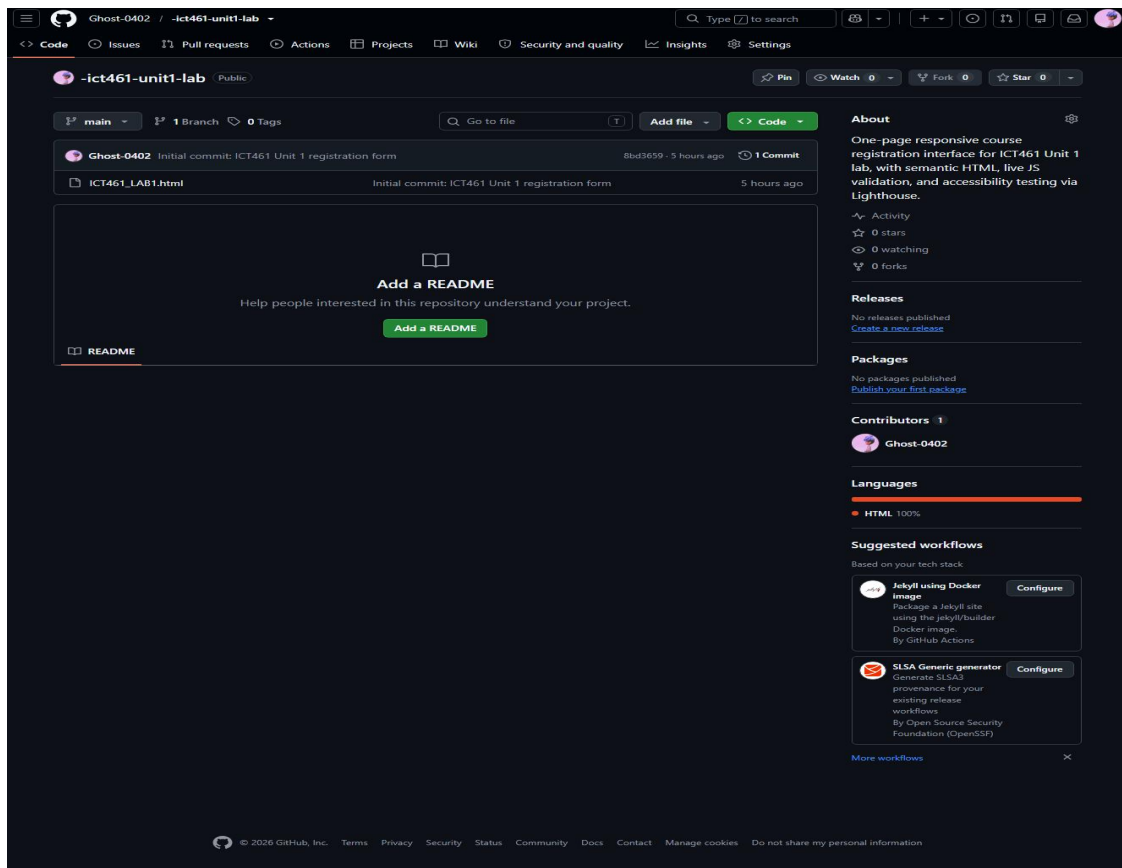
4.3 Re-Test Result



Purpose: I re-ran the Lighthouse accessibility audit after updating the CSS to confirm the contrast fix resolved the issue.

Result: The accessibility score improved from 19/20 to a perfect 20/20, with all checks passing.

5. Git History



Purpose: This screenshot shows my GitHub repository, confirming the project is under version control and hosted online.

What I did: I initialized a local Git repository with `git init`, staged and committed the completed HTML file, created a new empty repository on GitHub, linked it as a remote with `git remote add origin`, and pushed my commit using `git push -u origin main`.

Repository link: <https://github.com/Ghost-0402/-ict461-unit1-lab>

6. Reflection (300 words)

For this lab, I built a one-page course registration interface for ICT461 using semantic HTML, responsive CSS, and a JavaScript interaction. I structured the page with landmark elements such as header, main, and footer, and used `aria-labelledby` attributes to connect each section to its heading. The form includes labelled inputs for name, email, student ID, and course selection, plus a live “Registration Summary” panel that updates instantly as the user types, with no page reload required.

I paid close attention to validation and keyboard accessibility. Each field shows clear error messages using `aria-invalid` and `role="alert"`, so the form communicates problems to both sighted and screen reader users. All controls are native HTML elements, so keyboard navigation and focus states worked correctly without extra code.

Running a Lighthouse accessibility audit was the most useful part of this lab. My first score was 19/20, with the only failure being insufficient colour contrast on my muted Grey text. I had assumed the colour looked fine, but the audit showed it did not meet WCAG contrast requirements. I fixed this by darkening the CSS variable from `#6b7280` to `#4b5563`, and re-running the audit confirmed a perfect 20/20 score. This taught me that accessibility issues are not always obvious from a visual check, and that automated tools catch things I would otherwise miss.

Using DevTools to inspect the Elements and Network tabs helped me understand how the browser actually renders and loads my page, rather than just trusting my code “looked right” in the editor. Setting up Git and GitHub was also a good refresher on version control: initializing a repository, committing with clear messages, and pushing to a remote so the project is properly tracked and shareable.

If I continued this project, I would split the CSS and JavaScript into separate files and test with a real screen reader rather than relying only on automated audits.